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應用科學學院
Faculdade de Ciências Aplicadas
Faculty of Applied Sciences

Doctor of Philosophy in Computer Applied Technology

Thesis Defence Examination

Thesis Review Comments

Student

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Title of Thesis	Improving the Reliability and Accuracy of Dialogue-Based Natural Language Generation		
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Reviewer

Name	Prof. Luísa Coheur	Date	2026-06-03
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Recommendation

☒ Accepted ☐ Suggestion for Revision*

* "Suggestion for Revision" will result in the delay of the oral defence.

Comments

Overview.

This thesis focuses on dialogue summarization, addressing the challenges of generating different types of coherent and faithful summaries from multi-turn conversational data. Chapters 2 and 3 cover the literature review and the mathematical foundations, respectively. Chapter 3 also includes the datasets and evaluation metrics commonly used in the thesis. Chapters 4, 5, and 6 present a coherent body of contributions of the thesis, by exploring three axes: role interaction, topic structure, and dynamic semantic adaptation.

In Chapter 4, the author introduces Role-Aware Adapters (RAA), a module that refines encoder representations before decoding by explicitly incorporating role interaction and semantic associations into the weighting process.

Chapter 5 proposes the Hierarchical Topic-Driven Approach for Role-Oriented Dialogue Summarization (HiSum), which models utterances in a latent semantic space and identifies more precise topic distributions. The resulting topic structure is then used to guide summary generation through hierarchical attention mechanisms.

In Chapter 6, sentence embeddings distributions are used to compute adaptive topic

weighting, producing semantically reweighted sentence representations. These are further combined with positional signals prior to decoding.

Chapter 7 introduces a Critical Scoring Model (SSCM), designed to reduce hallucinations in Chinese financial text generation. This component learns to evaluate and rank generated outputs. Contrastive training data is built for the training process. This provides an additional layer of quality control over the generated summaries.

Finally, Chapter 8 presents a comparative study between reasoning and non-reasoning large language models in the context of dialogue summarization. The results are particularly interesting, as they challenge some common assumptions in the field regarding the superiority of reasoning-based models for this task.

The thesis concludes in Chapter 9, which presents the main conclusions, discusses the limitations of the work, and outlines directions for future research.

Knowledge Base and Literature Review.

The literature review is clear, well-structured, and pedagogical. As stated in Chapter 2, it is organized around three closely related dimensions: generative paradigms, structural modeling, and mechanisms for improving reliability.

The review is comprehensive and covers the most relevant and recent work in dialogue summarization and related areas. It concludes with a well-motivated discussion of research gaps, which are clearly connected to the contributions developed throughout the thesis, providing a solid foundation for the proposed methods.

Originality and Independence.

The research explores multiple directions within dialogue summarization, each motivated by clearly identified gaps in the existing literature. The proposed methods go beyond straightforward adaptations of existing approaches, introducing novel architectural components and modeling strategies.

The work demonstrates a good level of scientific independence, as the author defines the research directions, designs the methodological contributions, and integrates multiple lines of investigation into a coherent research agenda, rather than relying solely on incremental improvements of existing models.

Arguments and Reasoning.

All conclusions are strongly supported by the experimental results. The reasoning is clear, logically structured, and well documented throughout the thesis. The author also discusses limitations of the proposed methods, showing a balanced and critical understanding of the results.

Evaluation within the Field.

The experimental methodology is sound and well documented, ensuring a high degree of reproducibility. Each study follows a consistent structure, including problem definition, methodological design, experimental setup, results, and discussion.

The evaluations typically include ablation studies and human evaluation. A case study (in each one of the “main” chapters) is also provided to illustrate the practical behaviour of the proposed methods.

Completeness and Coherence.

The thesis forms a highly coherent research program, despite addressing multiple directions. The contributions presented in Chapters 4, 5, and 6 focus on improving accuracy in dialogue summarization, while Chapters 7 and 8 extend the investigation toward hallucination reduction, and the role of reasoning in large language models.

Taken together, these components form a unified contribution centered on improving dialogue summarization systems. The progression of ideas across chapters is natural and well motivated, resulting in a strong overall narrative.

Conclusion.

Overall, the thesis represents a strong scientific contribution to the field of dialogue summarization. The work is well structured, methodologically solid, and supported by comprehensive experimental validation. I consider that it meets the required standards for academic defense and can proceed as it stands.

Additional Comments.